

NUSROT JAHAN

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Education

Bachelor of Science in Information and Communication Engineering

Bangladesh University of Professionals (BUP)

Nov. 2021 – Aug. 2025

Dhaka, Bangladesh

- CGPA: **3.84** out of 4.00
- IELTS: Overall Band **7.5** (L: 8.0, R: 8.0, W: 7.0, S: 6.5)

Higher Secondary Certificate (HSC)

Viqarunnisa Noon College, Bailey Road

2020

Dhaka, Bangladesh

- GPA: **5.00** out of 5.00 (Board Scholarship)

Secondary School Certificate (SSC)

Ideal School and College, Banasree Branch

2018

Dhaka, Bangladesh

- GPA: **5.00** out of 5.00 (Board Scholarship)

Research Interest

- Interpretable AI for critical decision-making
- Natural Language Processing
- Diffusion Interpretability
- Generative AI & Large Language Models (LLMs) in Clinical Practice

Research Experience

Undergraduate Thesis

Oct. 2024 – Aug. 2025

Thesis Title: Transformer-Based Suicide Risk Classification from Social Media Text Using Clinically Valid Relabeled and Custom-generated Dataset

Supervisor: Dr. Khondokar Habibul Kabir, Department of EEE, Islamic University of Technology (IUT)

- Designed a consensus-driven, dual-model agreement pseudo-labeling pipeline that integrates a traditional ML ensemble (SVM, RF, Gradient Boosting, LR) with a custom dual-attention transformer (mental-bert-base-uncased) to automatically generate the largest clinically aligned dataset (85,451 r/SuicideWatch posts) featuring C-SSRS-mapped four-tier risk labels (High/Moderate/Low/No Risk).
- Enhanced label reliability and mitigated severe class imbalance through cross-model consensus filtering, T5-based clinical paraphrasing, optimized loss functions, and confidence-ranked downsampling, substantially improving recall for rare high-risk categories.
- Finally, fine-tuned and optimized BERT, DistilBERT and RoBERTa for balanced accuracy and efficiency, enabling scalable real-time deployment in clinical/public health settings.

Publications

2024

- **Published:** Kazi Nusrat Jahan Suchi, **Nusrot Jahan**, Arpa Tasnim, Nazneen Akhter, “Explainable attention based BiLSTM-SVM for Software Requirement Classification: Integrating Generative Artificial Intelligence”, *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024. [Link]
- **Published:** Saif Rahman, Tarannum Binte Tariq, **Nusrot Jahan**, Naimul Hasan Naime, Nazneen Akhter, “HAN-DistilBERT for Depression Detection from Twitter Data: Integrating Hierarchical Attention with Transformer-Based Model”, *2024 27th International Conference on Computer and Information Technology (ICCIT)*, 2024. [Link]
- **Published:** Tarannum Binte Tariq, **Nusrot Jahan**, Saif Rahman, Naimul Hasan Naime “Enhanced Twitter-Based Depression Severity Classification Using Multi-Head Attention with RoBERTa and DistilBERT Fusion”, *7th International Conference On Sustainable Technologies For Industry 5.0 (STI)*, 2025. [Link]
- **Published:** Nusrat Yasmin, K. Habibul Kabir, Tasnim Fariha, **Nusrot Jahan** “Empowering Educators: A Machine Learning System for Predicting Student Mental Health in Bangladesh”, *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. [Link]
- **Published:** Jakaria Kamal, **Nusrot Jahan**, Jasiya Fairiz Raisa “A Lightweight and Resource-Aware Framework for Multilingual Cyberbullying Detection”, *2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. [Link]

- **Ready Manuscript: Nusrot Jahan**, Tasnim Fariha, Jasiya Fairiz Raisa, K. Habibul Kabir “A Transformer-Based Framework for Suicide Risk Detection on Social Media Leveraging Clinically Validated Data”
- **Ready Manuscript: Nusrot Jahan**, “Clinical Symptom-Based Multiclass Migraine Classification: Addressing Class Imbalance and Feature Heterogeneity Through Evaluation of Machine Learning Models”, prepared as part of “Machine Learning Laboratory” coursework.
- **Ready Manuscript: Kazi Nusrat Jahan Suchi, Nusrot Jahan**, Tarannum Binte Tariq, “Dual-Band Defective Ground Structured 1x4 Microstrip Array Antenna with S-Shaped Slots for 5G”, prepared as part of “Microwave Engineering and Antenna Laboratory” coursework.
- **Ready Manuscript: Tasnim Hossain Fariha, Nusrot Jahan**, Salman Jahan Rafi, “Design and Performance Analysis of Dual-Band Microstrip Patch Antenna for 5G Applications”, prepared as part of “Telecommunication Systems” coursework.

Work Experience

Junior Software Engineer

Mar. 2025 – Aug. 2025

Integrated Software and Technologies Limited (ISTL)

- Built enterprise-grade microservices systems using Spring Boot/Cloud, including a Product Management System (OAuth 2.0, PostgreSQL, Redis, Kafka, MinIO) and an aged care coordination platform with JWT-secured role-based APIs and structured Carer-Senior allocation workflows.
- Completed GIS training with hands-on experience across the full spatial data pipeline: QGIS styling, PostGIS integration, GeoServer WMS/WFS publishing, and OpenLayers-based live map visualization.

Intern

Nov. 2024 – Feb. 2025

Integrated Software and Technologies Limited (ISTL)

- Developed a fully secure operational E-Visa mobile application to streamline visa processing through advanced authentication mechanisms included integrating biometric features such as facial recognition and liveness checking, fingerprint extraction, Email OTP sending and verification.
- NID text extraction using various OCR (Optical Character Recognition) tools or libraries.

Projects

1. Coffee Shop Sales Analysis

Nov. 2024

- Developed an interactive Streamlit dashboard to analyze sales data (Jan–Jun 2023) from multiple coffee shop branches.
- Implemented dynamic filters (date range, store location, product category) for interactive data exploration and insights.
- Visualized key sales metrics including total revenue, quantity sold, top-performing products, and location-based trends.
- **Tools:** Streamlit, Pandas, Plotly, Matplotlib, Seaborn, Altair, Pillow — [Project Link](#)

2. Multiple-Diseases-Prediction-using-AI

May 2024

- Developed a Streamlit-based web application for predicting multiple diseases, including diabetes, Parkinson’s disease, and heart disease, using machine learning algorithms.
- Deployed predictive models using Streamlit for interactive, user-friendly visualization.
- **Tools:** Python3, Jupyter Notebook — [Project Link](#)

3. E-learning Platform with Two-way Attendance System

May 2024

- Designed and implemented an education platform including an admin panel for managing teachers and students.
- Built features for student/teacher profile management and automated two-way attendance tracking- allowing teachers to mark attendance through radio-button selection from the list of students as well as enabling students to record their own attendance via OTP verification.
- **Tools:** CSS, HTML5, PHP, JavaScript, XAMPP, MySQL — [Project Link](#)

4. Product Management System Website

Apr. 2024

- Developed a dynamic product management system with a rotating slider to showcase trending products.
- Implemented product validation for data integrity and improved user experience with enhanced UI.
- **Tools:** CSS, HTML5, PHP, JavaScript, XAMPP, MySQL — [Project Link](#)

5. Client Server Platform for Real-Time Chat, File Transfer and Multiplying Game

Apr. 2022

- Developed a Java-based client-server chat application with group chat functionality.
- Integrated gaming features and seamless file transfer for real-time interaction.
- **Tools:** Java, MySQL — [Project Link](#)

Skills

Programming Languages: C, C++, Python, Java, PHP, Dart

Database Management: PostgreSQL, MySQL, Redis, MinIO (Object Storage), PostGIS

Machine Learning and AI: TensorFlow, PyTorch, Transformer Architectures (Encoder, Decoder, Vision)

Computer Vision & Biometrics: OpenCV, Mediapipe Vision/Face Landmarker, DeepFace, OCR (Pytesseract, EasyOCR, Google Cloud Vision), Fingerprint Processing (WSQ, MinIO integration)

Version Control: Git, GitHub

Development Tools: Visual Studio Code, Android Studio, IntelliJ IDEA, Eclipse IDE, Docker, Google Colab, Kaggle Notebooks

Project Management Tools: Jira, Trello

Simulation & Modeling: CST Studio Suite (Electromagnetic Simulation), MATLAB

Productivity Software: MS Office Suite, Canva, LaTeX (Overleaf)

Soft Skills: Problem-solving, Analytical Thinking, Attention to Detail, Teamwork

Languages: English(Fluent), Bengali(Native)

Awards & Honors

1. 15th as a team among contestants from 40 universities of all around Bangladesh in the **1st Data Innovators Challenge** (October, 2024)
2. 2nd Runner-up in the **Research Based Poster Presentation** hosted by IEEE BUP SB, WIE Affinity Group & Computer Society (July, 2023)
3. 2nd Runner-up in the **National Entrepreneurship Summit, 2022**
4. Achieved **University Scholarship** based on merit (Year 3rd- Semester 2nd)

References

1. Dr. Khondokar Habibul Kabir

Undergraduate Thesis Supervisor

Professor, Department of Electrical and Electronic Engineering
Islamic University of Technology
Email: habib@iut-dhaka.edu

2. Jasiya Fairiz Raisa

Undergraduate Thesis Co-Supervisor

Lecturer, Department of Information and Communication Engineering
Bangladesh University of Professionals
Email: jfraisa@bup.edu.bd